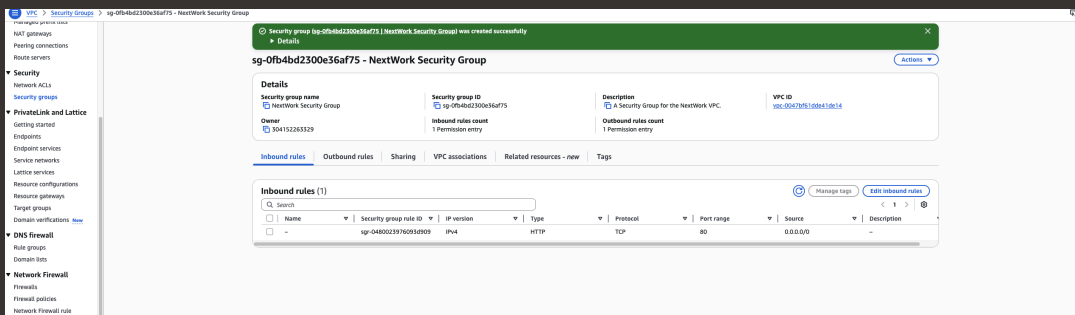




VPC Traffic Flow and Security

JJ

jjoseph1608@outlook.com





Introducing Today's Project!

What is Amazon VPC?

Amazon Virtual Private Cloud (Amazon VPC) gives you full control over your virtual networking environment, including resource placement, connectivity, and security.

How I used Amazon VPC in this project

Within the VPC I Set up route tables, Implemented security groups and deployed network ACLs

One thing I didn't expect in this project was...

How convenient and easy it was to setup

This project took me...

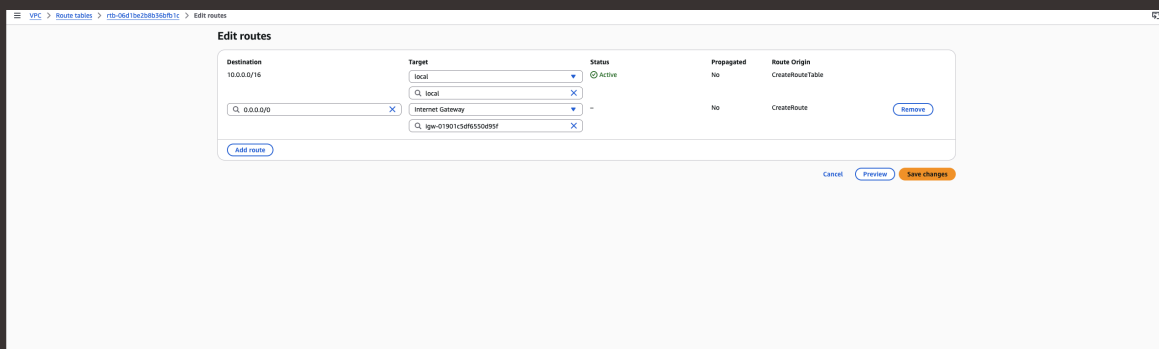
30 minutes



Route tables

Route tables are where the data in your network should go. Every subnet in your VPC needs to be linked to a route table because the table tells your subnet's traffic where to travel to send and receive data. For example, if you have a web server

Route tables are needed to make a subnet public because it contains the rules that determine where network traffic is directed

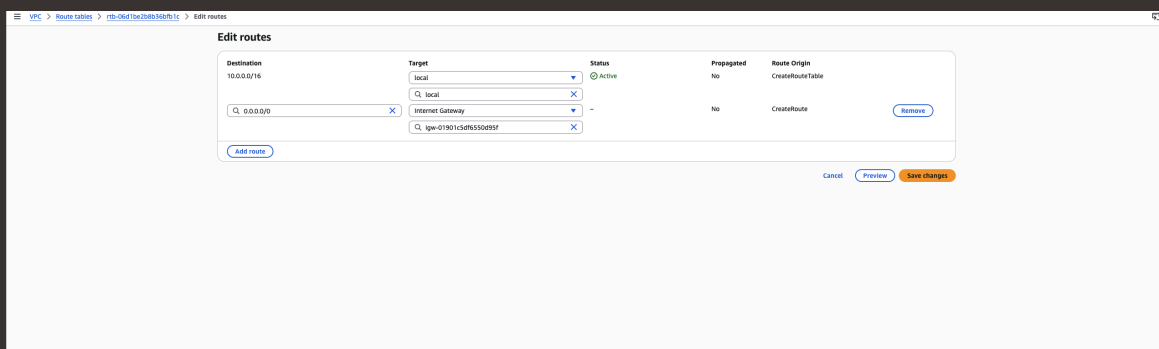




Route destination and target

Routes are defined by their destination and target, which mean a route's destination is the range of IP addresses that traffic should be sent to, and the target is the network device that will send the traffic there

The route in my route table that directed internet-bound traffic to my internet gateway had a destination of all IP addresses and a target of the public internet





Security groups

A security group is a security checkpoint. Security groups are responsible for checking who comes in and out. They have strict rules about what kind of traffic can enter or leave the resource based on its IP address, protocols and port numbers.

Inbound vs Outbound rules

Inbound rules control the data that can enter the resources in your security group, while outbound rules control that data that your resources can send out.

An outbound rule controls the traffic that leaves a system or network to the internet or network. My security groups outbound rule allows all outbound traffic via AWS' default setting.



Security group sg-0fb4bd2300e36af75 | NextWork Security Group

Details

Actions

Details

Security group name

NextWork Security Group

Security group ID

sg-0fb4bd2300e36af75

Description

A security group for the NextWork VPC.

VPC ID

vpc-05a7379f5160e418e14

Owner

304152263329

Inbound rules count

1 Permission entry

Outbound rules count

1 Permission entry

Inbound rules

Outbound rules

Sharing

VPC associations

Related resources - new

Tags

Inbound rules (1)

Manage tags

Edit inbound rules

Name	Security group rule ID	IP version	Type	Protocol	Port range	Source	Description
-	sg-04800239760954909	IPv4	HTTP	TCP	80	0.0.0.0/0	-



Network ACLs

Network ACLs are rules that either allow access to a computer environment or deny it. In a way, an ACL is like a guest list at an exclusive club. Only those on the list are allowed in the doors.

Security groups vs. network ACLs

The difference between a security group and a network ACL is that security groups allow you to control inbound and outbound traffic at the instance level, while network ACLs offer similar capabilities at the VPC subnet level.

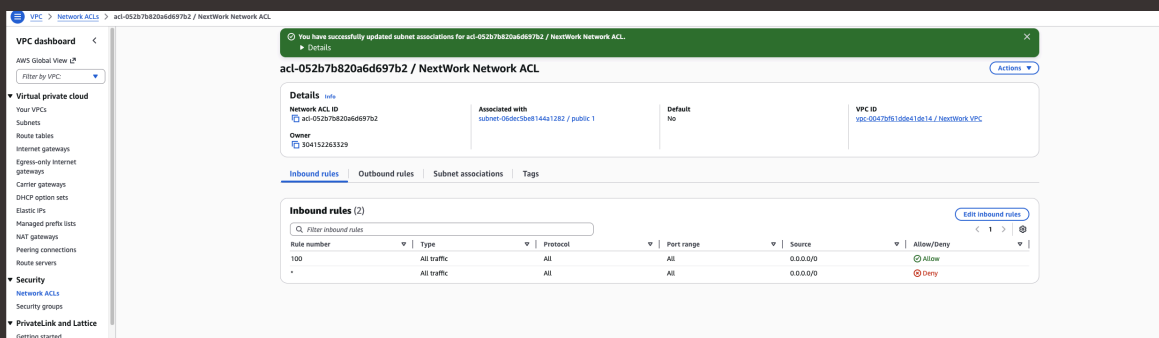


Default vs Custom Network ACLs

Similar to security groups, network ACLs use inbound and outbound rules

By default, a network ACL's inbound and outbound rules will allow all traffic inbound and outward.

In contrast, a custom ACL's inbound and outbound rules are automatically set to deny all traffic until you add rules about the kind of traffic we allow



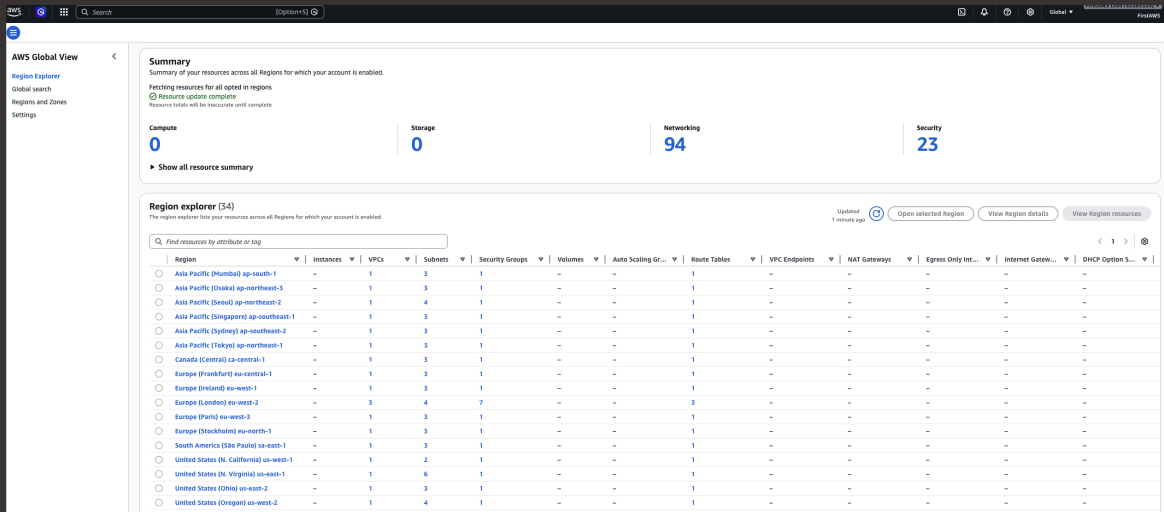


Tracking VPC Resources

I created additional VPC resources in a new region using the AWS CLI. Instead of my usual region (Europe, London, eu-west-2), I used US East (N. Virginia, us-east-1). Teams would use multiple regions to reduce latency for users in different parts of the world, improve reliability by spreading resources across locations, and ensure their applications stay available even if one region experiences an outage.

EC2 Global View is a tool where you can find all of your AWS resources across every region in your account in one place, including VPCs, subnets, instances, security groups and internet gateways. I could even narrow down my search by filtering by resource type or region to find exactly what I am looking for. Without EC2 Global View, you would have to manually switch between each region one by one in the console to track down your resources, which would be time consuming and easy to lose track of.

Now that I have learnt about EC2 Global View, I would use it again to get a quick overview of all my resources across every region, especially when troubleshooting, cleaning up unused resources to avoid unexpected charges, or working on projects that span multiple regions and need everything tracked from one central place.





nextwork.org

The place to learn & showcase your skills

Check out nextwork.org for more projects

